

← History

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❑ clc;
clear;
close all;
% GSM TDMA Parameters
Frame_Duration = 4.615;           % GSM frame
duration in ms
Total_Slots = 8;                  % Number of time
slots in a GSM frame
Active_Users = 6;                 % Number of
active users
% Objective 1: Calculate Time Slot Duration
TimeSlot_Duration = Frame_Duration /
Total_Slots;
% Objective 2 & 3: Allocate Time Slots
Allocated_Slots = min(Active_Users,
Total_Slots);
Frame_Utilization = (Allocated_Slots /
Total_Slots) * 100;
% Display Results
disp('GSM TDMA System Performance');
disp('-----');
disp('--');
fprintf('Total Frame Duration      : %.3f
ms\n', Frame_Duration);
fprintf('Total Time Slots           : %d\n',
Total_Slots);
fprintf('Time Slot Duration           : %.4f
ms\n', TimeSlot_Duration);
fprintf('Active Users                 : %d\n',
Active_Users);
fprintf('Allocated Time Slots          : %d\n',
Allocated_Slots);
fprintf('Frame Utilization              : %.2f %%
\n', Frame_Utilization);
% Time Slot Allocation
Slot = zeros(1, Total_Slots);
for i = 1:Allocated_Slots
    Slot(i) = i;
end
% Plot 1: Time Slot Allocation
figure;
stem(1:Total_Slots, Slot,
'filled','LineWidth',2);
title('GSM TDMA Time Slot Allocation');
xlabel('Time Slot Number');
ylabel('User Number');
grid on;
% Plot 2: Frame Utilization
figure;
bar([Allocated_Slots
Total_Slots-Allocated_Slots]);
set(gca,'XTickLabel',
{'Allocated','Unallocated'});
ylabel('Number of Slots');
title('Frame Slot Utilization');

```

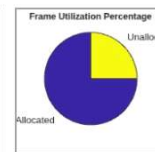
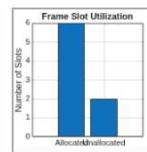
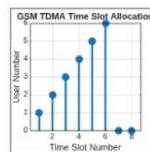
10:39 ①

VoLTE 4G+  



GSM TDMA System Performance

Total Frame Duration	: 4.615 ms
Total Time Slots	: 8
Time Slot Duration	: 0.5769 ms
Active Users	: 6
Allocated Time Slots	: 6
Frame Utilization	: 75.00 %



>> Enter command





Figure 1: clc; clear; close all; % GSM TDMA Parameters Frame_D...

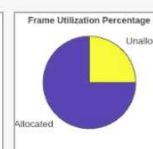
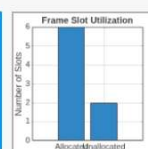
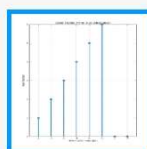
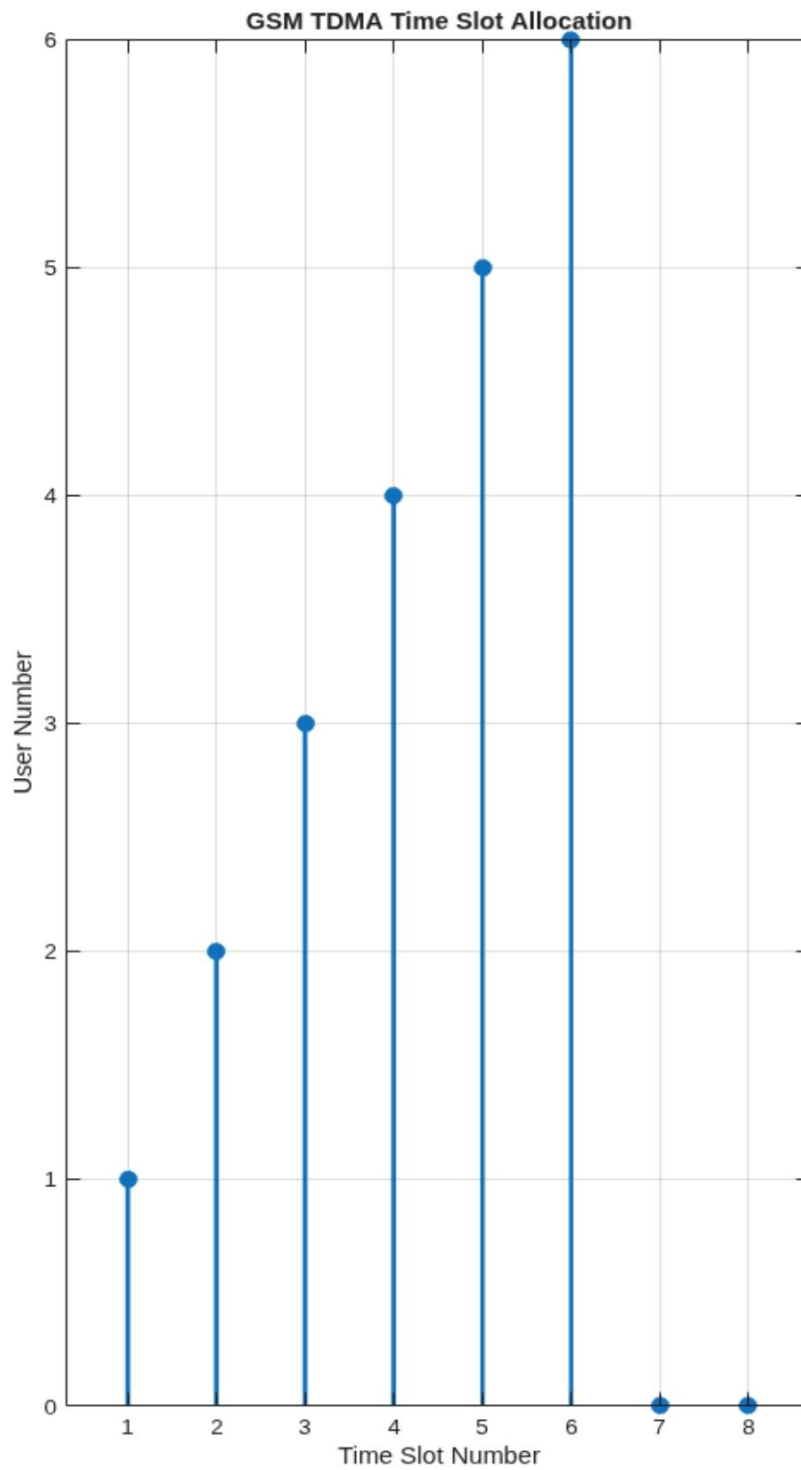




Figure 2: clc; clear; close all; % GSM TDMA Parameters Frame_D...

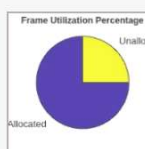
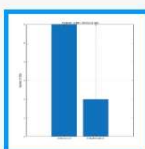
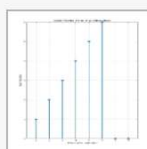
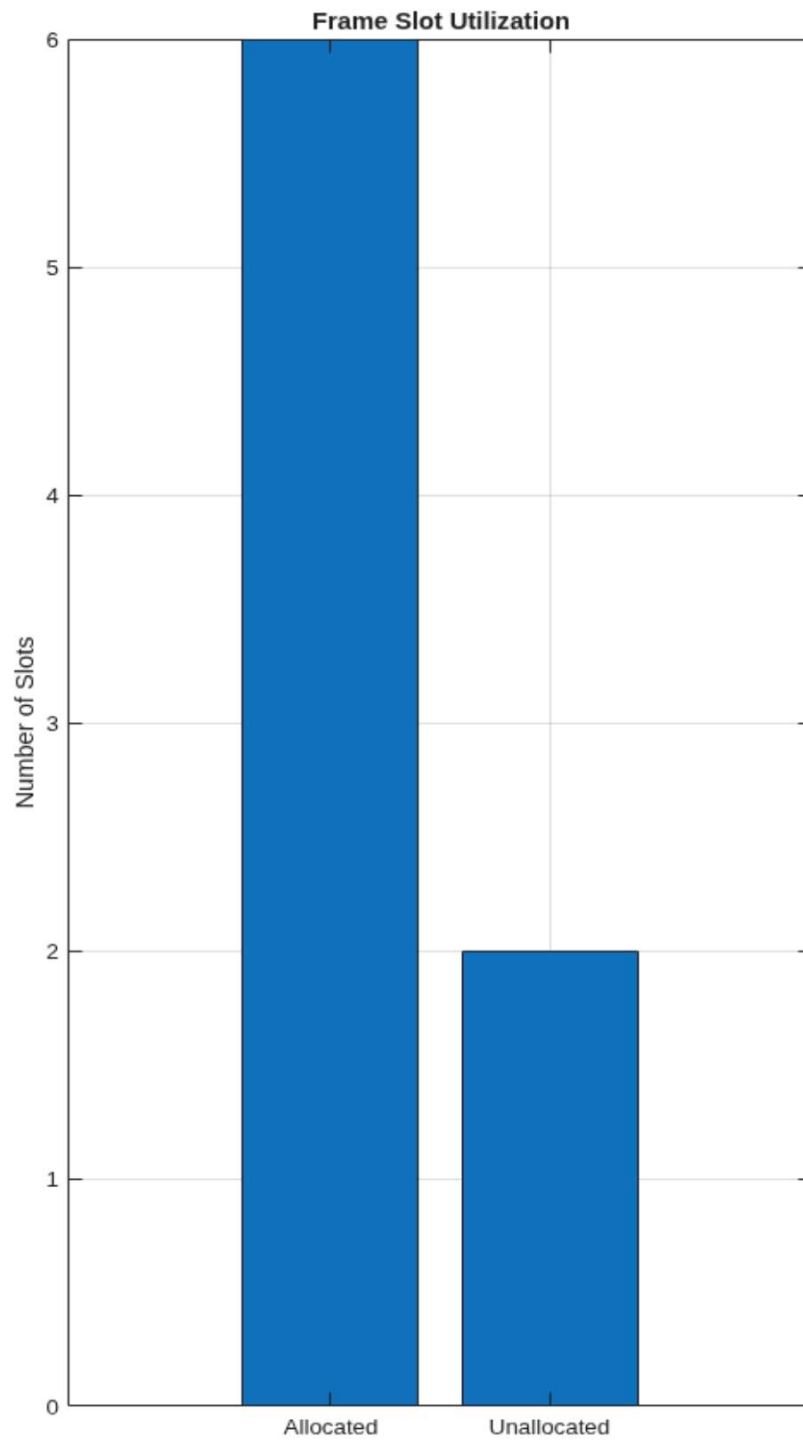




Figure 3: clc; clear; close all; % GSM TDMA Parameters Frame_D...

